EECS 3311 Software Design - Fall 2026 - Course Project

Stage 2 Implementation Report

AI Study and Learning Agent

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Repository	https://github.com/Nazaninsoleimani/EECS3311 (project folder `AI Study and Learning Agent/`)
Stage 1 design	`docs/stage1/Stage1_Report.pdf`
Implementation period	October 20 - November 10, 2026
Language and build	Java 21, Gradle multi-module (`backend`, `frontend:cli`, `frontend:gui`)
LLM	Google Gemini (`gemini-2.5-flash`, free tier) through the `LLMClient` adapter; any OpenAI-compatible model by configuration

How to read this report: section 2 traces every Stage 1 feature (F01-F16) to the classes, methods, patterns and demonstration video that realise it; section 3 explains the six design patterns with the updated class diagrams; section 4 lists every difference from the Stage 1 design with its reason; sections 5-7 cover AI collaboration, commits and verification, and the video index. Items highlighted like this are completed by the student before submission.

1. Project overview and implementation status

Application. The AI Study and Learning Agent is a desktop application for university students. A student creates a course, uploads their own lecture notes (PDF, DOCX, TXT, Markdown), and the agent works over that material: it answers questions with citations to the source pages, summarises lectures, extracts the course's concepts and their relations, plans study time up to the exam, generates and grades quizzes and flashcards, finds weak topics, adapts practice difficulty, and tracks progress. Everything is available through a Swing GUI (12 views) and a `study` command-line interface (32 commands); both call the same `LearningFacade`.

Users. A single student per installation (a local profile, `STUDENT_ID`, default `student-1`). All data is stored locally in one SQLite file (`~/.study-agent/study-agent.db`).

AI / LLM integration and agent behaviour. The LLM is never called directly by a feature. Every AI goal goes through `LearningAgent`, which (1) loads the learner context from memory (mastery, recent history, exam date), (2) asks the `Planner` for a plan of tool steps, (3) executes the steps through the `ToolRegistry` (retrieval, progress analysis, scheduling, generation, reminders), letting the Planner revise the remaining plan after each observation (for example, no generation when retrieval found nothing), (4) builds prompts with retrieved sources inside a token budget (`ContextBuilder`), and (5) validates every LLM output (`ResponseValidator`: JSON schema fields, citation grounding) before it becomes a domain object, regenerating once when validation fails. Session memory keeps the tutor conversation so follow-up questions are rewritten into standalone retrieval queries. Deterministic code makes the decisions that must be reliable (mastery, difficulty transitions, scheduling strategy, grading of objective questions); the LLM generates content.

Working functionality. All 16 Stage 1 features are implemented and working in both front ends (table below). The build compiles three modules and runs 118 automated JUnit tests, all passing (section 6). Without an API key the deterministic features work fully and AI features use documented fallbacks (extractive answers, deterministic plan priorities, stored questions) or report clearly that the AI is unavailable.

ID	Feature	Type	Status	GUI view	CLI command
F01	Course Management	Deterministic	Implemented	Courses	`course create|list|update|delete`

F02	Material Upload and Management	Deterministic	Implemented (modified)	Materials	`material upload|list|delete|reindex`
F03	Document Summarization	AI-based	Implemented	Materials	`material summarize`
F04	Concept Extraction	AI-based	Implemented	Materials	`material extract-concepts`, automatic on upload
F05	Question Answering over Materials	Hybrid	Implemented (modified)	Tutor	`ask`, `ask -i`
F06	Personalized Study-Plan Generation	Hybrid	Implemented	Study Plan	`study-plan generate|show|done`
F07	Flashcard Generation and Review	Hybrid	Implemented	Flashcards	`flashcards generate|list|review`
F08	Quiz Generation	AI-based	Implemented	Quiz	`quiz generate|list`
F09	Automatic Quiz Grading	Hybrid	Implemented	Quiz	`quiz take`, `quiz generate --take`
F10	Answer Explanation	AI-based	Implemented	Quiz (Explain)	`quiz explain`
F11	Weak-Topic Identification	Hybrid	Implemented	Progress	`weak-topics show`
F12	Adaptive Question Difficulty	Hybrid	Implemented (modified)	Learning Session	`learning-session start`
F13	Learning Progress Tracking	Deterministic	Implemented	Progress, Dashboard	`progress show`
F14	Study Reminder Management	Deterministic	Implemented	Reminders	`reminder add|list|cancel|dispatch`
F15	Concept Relationship Visualization	Hybrid	Implemented	Concept Map	`concept-map show [--format mermaid]`
F16	Learning Session History	Deterministic	Implemented	History	`history show`

Limitations and incomplete functionality (honest list).

- **Automated tests use a scripted LLM.** The 118 tests exercise every AI flow with a deterministic fake `LLMClient`, so they verify the agent's control flow, validation and fallbacks, not the quality of real model output. Live-model behaviour was checked manually (section 6.2) [confirm after your run with the Gemini key].
- **Retrieval quality.** The default embeddings (`LocalHashEmbeddingService`) are lexical (hashed, stemmed words), not semantic, so a question phrased with different words than the notes may retrieve less. Gemini embeddings can be enabled with `EMBEDDING_PROVIDER=gemini` (section 4).
- **Scanned PDFs** without a text layer are rejected with a clear reason; there is no OCR.
- **GUI details planned in Stage 1 that were simplified:** tutor citations are shown as text, not clickable links to the page; history entries do not open the related object (the reference id is stored); the material status badge is a table column; the concept map uses a circular layout rather than an interactive force layout.
- **Single local user.** There is no login or multi-user support (not required by Stage 1).
- **Short-answer grading by the LLM** can vary between runs; scores outside 0-1 or malformed output are rejected and the keyword grade is kept and excluded from mastery.
- Full deterministic testing and KUMA agent validation are Stage 3 work.

2. Feature to implementation mapping

All classes are under `ca.yorku.eecs3311.studyagent`. Paths are abbreviated: `app/` = `backend/.../application`, `agent/`, `tools/` = `agent/tools`, `domain/`, `integ/` = `integration`, `persist/` = `persistence`, `gui/` = `frontend/gui/.../presentation/gui`, `cli/` = `frontend/cli/.../presentation/cli`. Every feature is reached from the GUI view or CLI command through `LearningFacade` (Facade).

F01 - Course Management

Stage 1 reference	F01, UC01, SD11
Status	Implemented (Stage 1 rule "exam date in the past is rejected" enforced in week 3)
Classes / files and responsibilities	`gui/CourseView` (form, table, selection); `cli/CLIApplication` (course commands); `app/LearningFacade` (entry, history event); `app/CourseService` (unique code per student, exam-date check, cascading delete, resolve by id or code); `domain/Course`; `persist/CourseRepository` + `SqliteCourseRepository`; `agent/MemoryManager` (records `COURSE_*` events)
Key methods	`createCourse()`, `updateCourse()`, `deleteCourse()`, `listCourses()`, `resolve()`, `requireUniqueCode()`, `requireFutureExam()`
Design patterns	Facade
Video	[link + timestamp]

Deterministic The view or CLI passes code, title and exam date to `LearningFacade.createCourse()`; `CourseService` builds the `Course` (which normalises the code), rejects a duplicate code or a past exam date, and saves it; the facade records `COURSE_CREATED`. Deleting asks for confirmation in both front ends, then removes the course's materials, vectors, concepts, progress, quizzes, decks and plans.

F02 - Learning Material Upload and Management

Stage 1 reference	F02, UC02 (includes UC04), SD07
Status	Implemented, modified: upload returns `MaterialUpload` and runs concept extraction (UC04); embedding failures keep the material pending and indexing is retried (section 4, C1 and C3)
Classes / files and responsibilities	`gui/MaterialView` (file chooser, status table, Retry indexing); `app/MaterialService` (validation, parse, chunk, index, pending retry); `integ/parsing/DocumentParser` with `PdfParser` (PDFBox, per page), `DocxParser`, `PlainTextParser`; `integ/parsing/Chunker` (800-token chunks, 100 overlap, page refs); `integ/retrieval/RetrievalService`, `EmbeddingService`, `SqliteVectorStore`; `domain/CourseMaterial`, `DocumentChunk`; `app/MaterialUpload`
Key methods	`uploadMaterial()`, `upload()`, `findParser()`, `supports()`, `parse()`, `chunk()`, `index()`, `embed()`, `upsert()`, `markIndexed()`, `markFailed()`, `markIndexingPending()`, `retryPendingIndexing()`
Design patterns	Facade; parsers are interchangeable `DocumentParser` implementations
Video	[link + timestamp]

Deterministic `MaterialService.upload()` checks the file exists, its type has a parser and it is under 50 MB, then parses, chunks and saves the chunks, and indexes them (embed + upsert). Unsupported types are rejected before anything is stored; empty or unreadable files are stored as `FAILED` with the reason; an embedding outage leaves the material `UPLOADED` with its chunks so `retryPendingIndexing()` (run at start-up, `material reindex`, Retry indexing button) can finish later.

F03 - Document Summarization

Stage 1 reference	F03, UC03, SD05
Status	Implemented
Classes / files and responsibilities	`gui/MaterialView` (length choice, summary panel); `app/LearningFacade.summarizeMaterial()` (requires an indexed material, stores the summary); `agent/LearningAgent.summarize()`; `agent/Planner` (chooses single pass or map-reduce above 6,000 tokens); `tools/SummarizationTool` (map: one LLM call per group of chunks; reduce: structured JSON); `agent/ResponseValidator`; `domain/Summary`, `SummaryOptions`
Key methods	`summarizeMaterial()`, `summarize()`, `createPlan()`, `execute()`, `mapSections()`, `generate()`, `generateStructured()`, `validate()`, `saveSummary()`
Design patterns	Facade, Adapter
Video	[link + timestamp]

AI-based The agent counts the material's tokens, the Planner picks the mode, and `SummarizationTool` produces `{summary, keyPoints, sourceRefs}`; invalid output is retried once; an LLM failure returns "Summary unavailable, retry later" and nothing partial is stored.

F04 - Concept Extraction

Stage 1 reference	F04, UC04 (included by UC02), SD07
Status	Implemented (runs automatically after upload since week 3, and on demand)
Classes / files and responsibilities	`app/LearningFacade.extractConcepts()` and `uploadMaterial()`; `agent/LearningAgent.extractConcepts()`; `tools/ConceptExtractionTool` (batches of chunks, reuses known course concepts, drops relations to unknown concepts and self-relations); `app/ConceptMapService.save()` (persists and remaps relation ids to stored concepts); `domain/Concept`, `ConceptRelation`, `ConceptGraph`; `persist/ConceptRepository` (merge by normalised name)
Key methods	`extractConcepts()`, `execute()`, `generateStructured()`, `validate()`, `saveAll()`, `saveRelations()`
Design patterns	Facade, Adapter
Video	[link + timestamp]

AI-based with deterministic validation. The LLM proposes concepts and typed relations (`PREREQUISITE_OF`, `PART_OF`, `RELATED_TO`) per batch; validation merges duplicates and discards invalid relations, reporting how many were dropped. If the LLM is unavailable the upload still succeeds and the student is told extraction can be re-run.

F05 - Question Answering over Course Materials

Stage 1 reference	F05, UC05 (includes UC17), SD02
Status	Implemented, modified: with no LLM the agent returns the most relevant passages with citations instead of an error (section 4, C5)
Classes / files and responsibilities	`gui/TutorChatView` (conversation, "Show agent plan"); `cli ask` and `ask -i`; `LearningAgent.answerQuestion()`; `MemoryManager`, `SessionMemory` (last 20 turns); `Planner.retrievalQuery()` (rewrites follow-ups); `tools/RetrievalTool` → `RetrievalService` (threshold); `ContextBuilder` (numbered sources, token budget); `ResponseValidator.checkGrounding()`; `domain/Answer`
Key methods	`askQuestion()`, `answerQuestion()`, `generateGroundedAnswer()`, `extractiveAnswer()`, `retrieve()`, `build()`, `checkGrounding()`, `appendToSession()`
Design patterns	Facade, Adapter
Video	[link + timestamp]

Hybrid Retrieval and the grounding check are deterministic; the LLM writes the answer only from the numbered sources. If no chunk passes the similarity threshold, the LLM is not called and the student is told the material does not cover the question. An answer whose citations are missing or point to non-existent sources is regenerated once with a stricter instruction; if it still fails it is returned with `grounded=false` and a warning.

F06 - Personalized Study-Plan Generation

Stage 1 reference	F06, UC06 (includes UC11, UC17), SD01, SD04
Status	Implemented (ungrounded-plan warning and one retry of invalid priorities added in week 3)
Classes / files and responsibilities	`gui/StudyPlannerView`; `cli study-plan`; `LearningAgent.generateStudyPlan()`, `prioritize()`, `revisePlan()`; `Planner` (analyze → retrieve → schedule → remind); `tools/ProgressAnalysisTool` (deterministic ranking); `tools/StudyScheduleTool` (Strategy context); `ExamCramStrategy`, `SpacedRepetitionStrategy`, `BalancedReviewStrategy`; `tools/ReminderTool`; `app/StudyPlanMonitor` (revision); `domain/StudyPlan`, `StudyTask`, `TopicPriority`
Key methods	`generateStudyPlan()`, `createPlan()`, `rankTopics()`, `prioritize()`, `selectStrategy()`, `buildSchedule()`, `revisePlan()`, `needsRevision()`, `completeStudyTask()`
Design patterns	Strategy, Observer (revision), Facade, Adapter
Video	[link + timestamp]

Hybrid `ProgressAnalysisTool` ranks concepts by priority = 0.6(1 - mastery) + 0.25 recency gap + 0.15 prerequisite weight; focus topics are boosted. The LLM may reorder the candidates and write a learning objective per topic and the rationale (names it invents are ignored; invalid output is retried once, then the deterministic ranking is used with a warning). The tool picks the strategy from the days left (≤ 7 cram, ≥ 21 spaced repetition, otherwise or without an exam balanced) and allocates 30-minute slots. Reminders are created for the first seven planned days. When a concept's mastery drops sharply or below 50% and the plan does not revisit it within three days, `StudyPlanMonitor` asks the agent for a revised plan and archives the old one.

F07 - Flashcard Generation and Review

Stage 1 reference	F07, UC07, SD08
Status	Implemented
Classes / files and responsibilities	`gui/FlashcardView` (generate, flip, Again/Hard/Good/Easy); `cli flashcards generate|list|review` (`CommandParser`, `StudyCommands`); `LearningAgent.generateFlashcards()`; `tools/FlashcardGenerationTool` (drops empty and duplicate cards); `domain/FlashcardDeck`, `Flashcard` (Leitner boxes); `LearningFacade.reviewFlashcard()` → `ProgressTracker.recordResult()`
Key methods	`run()`, `parse()`, `dispatch()`, `generateFlashcards()`, `execute()`, `saveDeck()`, `dueFlashcards()`, `reviewFlashcard()`, `recordReview()`, `toGradeResult()`
Design patterns	Facade, Adapter, Observer (review results feed mastery)
Video	[link + timestamp]

Hybrid The LLM writes front/back pairs only from retrieved material (no matching material: an error, nothing generated); a shortfall is reported on the deck. Review is deterministic: Good/Easy move the card up a box, Again/Hard back to box 1; intervals 1, 2, 4, 8, 16 days. Each rating is also a graded result for the card's concept.

F08 - Quiz Generation

Stage 1 reference	F08, UC08, SD03 part A
Status	Implemented
Classes / files and responsibilities	`gui/QuizView` (count, difficulty, types, concepts); `cli quiz generate`; `LearningAgent.generateQuiz()`; `tools/QuizGenerationTool` (Factory Method client, up to three rounds); `QuestionFactory` + `MultipleChoiceQuestionFactory`, `TrueFalseQuestionFactory`, `ShortAnswerQuestionFactory`; `domain/Question` and subclasses, `QuestionSpec`, `Quiz`; `persist/QuizRepository`
Key methods	`generateQuiz()`, `execute()`, `toSpec()`, `resolveConcept()`, `build()`, `createQuestion()`, `save()`
Design patterns	Factory Method, Facade, Adapter
Video	[link + timestamp]

AI-based The LLM returns question specifications as JSON; each is built by the factory registered for its type, which rejects invalid specs (wrong option count, unmatched answer, unknown concept, empty prompt). Rejection reasons are fed back for replacement questions; a quiz with fewer questions carries a notice. A whole-course quiz targets the course's known concepts so every answer updates mastery.

F09 - Automatic Quiz Grading

Stage 1 reference	F09, UC09, SD03 part B
Status	Implemented
Classes / files and responsibilities	`gui/QuizView` + `QuestionPanel` (inputs per type, feedback); `cli quiz take`; `LearningFacade.submitQuiz()`; `app/GradingService`; `Question.grade()` in each subclass; `domain/QuizAttempt`, `GradeResult`; `app/ProgressTracker.recordAttempt()`; `ConceptMastery`, `PerformanceRecord`
Key methods	`submitQuiz()`, `grade()`, `gradeResponse()`, `Question.grade()`, `gradeWithRubric()`, `computeScore()`, `recordAttempt()`, `applyResult()`, `notifyObservers()`
Design patterns	Observer, Facade, Factory Method products (polymorphic `grade()`)
Video	[link + timestamp]

Hybrid Multiple-choice and true/false are graded exactly by the question objects; unanswered questions score 0. Short answers are scored by the LLM against the model answer and rubric (validated 0-1 score); without an LLM the keyword grade is shown and flagged "needs review", which is stored but not applied to mastery. The attempt is saved and `ProgressTracker` updates mastery and notifies observers.

F10 - Answer Explanation

Stage 1 reference	F10, UC10 (extends UC09), SD03 part C
Status	Implemented (CLI also accepts the question number)
Classes / files and responsibilities	`gui/QuizView` (Explain button per question); `cli quiz explain`; `LearningFacade.explainAnswer()`; `LearningAgent.explainAnswer()`; `RetrievalTool`, `ContextBuilder`; `domain/Explanation`
Key methods	`requestExplanation()`, `explainAnswer()`, `findAttempt()`, `execute()`, `build()`, `generate()`
Design patterns	Facade, Adapter
Video	[link + timestamp]

AI-based The explanation is generated from the question, the correct answer, the student's answer and retrieved sources, citing them as [n]; it is marked not source-backed when it cites none. Without an LLM the stored correct answer, the grading feedback and the most relevant passage are shown.

F11 - Weak-Topic Identification

Stage 1 reference	F11, UC11, SD06
Status	Implemented ("at least 5 graded answers" threshold aligned in week 3)
Classes / files and responsibilities	`gui/ProgressDashboardView` (Analyze weak topics); `cli weak-topics show`; `LearningAgent.diagnoseWeakTopics()`, `diagnoseWithLlm()`; `tools/ProgressAnalysisTool.rankTopics()`; `domain/WeakTopicReport`, `WeakTopic`, `ConceptMastery`
Key methods	`analyzeWeakTopics()`, `diagnoseWeakTopics()`, `rankTopics()`, `recommendedActivity()`, `generateStructured()`, `validate()`
Design patterns	Facade, Adapter
Video	[link + timestamp]

Hybrid Deterministic: a concept is weak when its mastery is below 50% after at least three attempts; with fewer than five graded answers in the course the report says the data is insufficient and suggests where to start. With enough data the LLM reads the student's wrong answers and the relevant material and writes the likely misconception and a recommended activity; on failure the deterministic report is returned.

F12 - Adaptive Question Difficulty

Stage 1 reference	F12, UC12, SD04
Status	Implemented, modified: `submitActivityResult()` returns `ActivityOutcome`; stored-question fallback added (section 4, C2 and C4)
Classes / files and responsibilities	`gui/LearningSessionView`; `cli learning-session start`; `LearningFacade.startAdaptiveSession()`, `submitActivityResult()`, `endAdaptiveSession()`; `GradingService`; `ProgressTracker.recordActivity()`; `domain/AdaptiveLearningSession` (State context) with `FoundationState`, `IntermediateState`, `AdvancedState`; `LearningAgent.nextAdaptiveActivity()`; `QuizGenerationTool`; `app/QuestionBank`; `app/ActivityOutcome`
Key methods	`startAdaptiveSession()`, `submitActivityResult()`, `gradeResponse()`, `recordOutcome()`, `onResult()`, `setState()`, `preferredQuestionTypes()`, `nextAdaptiveActivity()`, `pickConcept()`, `QuestionBank.pick()`
Design patterns	State, Observer, Factory Method, Facade
Video	[link + timestamp]

Hybrid The state machine decides difficulty (three correct in a row: up one level; two wrong: down one level); the agent picks the most urgent concept not asked in the last two activities and asks the LLM for one question of a type preferred by the current state. If generation fails, an unused stored question from the course's earlier quizzes is reused (same level and preferred type first); with none left the session ends gracefully, and when every concept is mastered it suggests a mixed review quiz.

F13 - Learning Progress Tracking

Stage 1 reference	F13, UC13, SD06 (view), SD03/SD04 (updates)
Status	Implemented
Classes / files and responsibilities	`gui/ProgressDashboardView` (ProgressObserver; table with mastery bars, live update line), `gui/DashboardView` (summary); `cli progress show`; `app/ProgressTracker` (Subject), `ProgressObserver`; `domain/ConceptMastery` (EMA, alpha 0.3), `PerformanceRecord`, `LearningProgress`, `ProgressEvent`; `persist/ProgressRepository`
Key methods	`getProgress()`, `recordAttempt()`, `recordResult()`, `applyResult()`, `attach()`, `notifyObservers()`, `onProgressChanged()`, `loadProgress()`
Design patterns	Observer, Facade
Video	[link + timestamp]

Deterministic Quizzes, adaptive answers and flashcard reviews all go through `ProgressTracker`, which stores a performance record, updates the concept's mastery (m ← 0.7m + 0.3s) and notifies observers; the dashboard refreshes without the grading code knowing it exists. The empty state suggests a first activity.

F14 - Study Reminder Management

Stage 1 reference	F14, UC14, SD09
Status	Implemented
Classes / files and responsibilities	`gui/ReminderView`; `cli reminder add|list|cancel|dispatch`; `app/ReminderScheduler` (store, background dispatch every 60 s, ProgressObserver); `domain/Reminder` (recurrence); `integ/notification/NotificationGateway` with `DesktopNotificationGateway` (GUI, macOS notification / system tray) and `ConsoleNotificationGateway` (CLI); `tools/ReminderTool` (plan reminders)
Key methods	`scheduleReminder()`, `schedule()`, `cancel()`, `dispatchDue()`, `findDue()`, `isDue()`, `markDelivered()`, `send()`, `onProgressChanged()`
Design patterns	Observer, Facade
Video	[link + timestamp]

Deterministic One-off reminders in the past are rejected; daily and weekly ones move to their next occurrence after delivery; a failed delivery stays pending and is retried. Overdue reminders are delivered when the application next starts. Reminders are also created by the agent for planned days and by the scheduler when a concept falls below the weak threshold (once per concept).

F15 - Concept Relationship Visualization

Stage 1 reference	F15, UC15, SD10
Status	Implemented ("Quiz me on this" added in week 3)
Classes / files and responsibilities	`gui/ConceptMapView` (drawn graph coloured by mastery, details on click, Copy as Mermaid, Quiz me on this); `gui/GuiContext.requestQuizOn()` → `QuizView.prepareFor()`; `cli concept-map show [--format text|mermaid]`; `app/ConceptMapService.buildGraph()`, `renderText()`; `domain/ConceptGraph (neighbours()`, `prerequisitesOf()`, `toMermaid())`
Key methods	`loadConceptMap()`, `getConceptMap()`, `buildGraph()`, `selectConcept()`, `neighbours()`, `prerequisitesOf()`, `requestQuizOn()`
Design patterns	Facade
Video	[link + timestamp]

Hybrid Relations were proposed by the LLM during F04; viewing is deterministic and makes no LLM call. Grey = not practised, red = below 50%, amber = learning, green = at least 80%.

F16 - Learning Session History

Stage 1 reference	F16, UC16, SD11
Status	Implemented (entries are not clickable; see limitations)
Classes / files and responsibilities	`gui/HistoryView` (course, period and activity filters); `cli history show [--course] [--last 7d|2w] [--type]`; `LearningFacade` (records an event for every operation); `agent/MemoryManager.recordEvent()`, `getHistory()`; `domain/LearningEvent`, `HistoryFilter`, `LearningHistory`; `persist/LearningHistoryRepository`
Key methods	`loadHistory()`, `getHistory()`, `recordEvent()`, `append()`, `find()`, `matches()`
Design patterns	Facade
Video	[link + timestamp]

Deterministic 21 event types are recorded; the agent also reads the last two weeks of events as learner context when planning and answering.

3. Implemented design patterns

All six patterns proposed in Stage 1 are implemented as designed; none was replaced. Each solves a problem that appears in the running code, described below with the participating classes, files and methods.

3.1 Strategy - study scheduling

Design problem	Study time must be distributed differently depending on the time left before the exam; one method with all algorithms would grow a large conditional and be hard to test.
Participants and roles	Strategy: `StudyPlanningStrategy` (`buildSchedule(request, priorities)`). Concrete strategies: `ExamCramStrategy` (≤ 7 days: practice-heavy, weakest topics repeated, last day mixed review), `SpacedRepetitionStrategy` (≥ 21 days: revisits after 1, 3, 7, 14 days), `BalancedReviewStrategy` (otherwise or no exam date). Context: `StudyScheduleTool` (`selectStrategy()`, `execute()`).
Files	`backend/.../agent/tools/StudyPlanningStrategy.java`, `ExamCramStrategy.java`, `SpacedRepetitionStrategy.java`, `BalancedReviewStrategy.java`, `StudyScheduleTool.java`, shared slot arithmetic in `ScheduleSlots.java`
How it follows the pattern	The tool selects a strategy object from the request and delegates; the agent and plan code only see `StudyScheduleTool`. Each strategy is unit-tested with fixed priorities (`StudyStrategyTest`).
Matches Stage 1?	Yes. Thresholds (7 / 21 days) were fixed during implementation.

3.2 Factory Method - question creation

Design problem	LLM JSON specifications must become the right `Question` subclass with type-specific validation, without the generator switching on every concrete class.
Participants and roles	Creator: abstract `QuestionFactory` with template operation `build(spec)` (common checks) and factory method `createQuestion(spec, id)`. Concrete creators: `MultipleChoiceQuestionFactory`, `TrueFalseQuestionFactory`, `ShortAnswerQuestionFactory`. Product: `Question`; concrete products: `MultipleChoiceQuestion`, `TrueFalseQuestion`, `ShortAnswerQuestion`. Client: `QuizGenerationTool` (map of factories per `QuestionType`).
Files	`agent/tools/QuestionFactory.java` and the three factories, `QuizGenerationTool.java`; `domain/Question.java` and subclasses
How it follows the pattern	The client calls `factories.get(spec.type).build(spec)`; rejected specs raise `ValidationException` and are replaced. Both quiz generation (F08) and adaptive sessions (F12) create questions this way, and `GradingService` grades through the polymorphic `grade()`.
Matches Stage 1?	Yes.

3.3 Facade - one application API for GUI and CLI

Design problem	Both front ends must expose every feature; calling the agent, services and repositories directly would duplicate orchestration in two places and couple the UI to internal classes.
Participants and roles	Facade: `LearningFacade` (about 40 operations). Subsystem: `LearningAgent`, `CourseService`, `MaterialService`, `GradingService`, `ProgressTracker`, `ReminderScheduler`, `ConceptMapService`, `StudyPlanMonitor`, `QuestionBank`, `MemoryManager`, repositories. Clients: the 12 `BaseView` subclasses, `CLIApplication` and `StudyCommands`.

Files	`application/LearningFacade.java`; wired once in `config/AppContext.java`
How it follows the pattern	No view or CLI class imports a service, the agent or a repository; the facade only delegates and records the history event. Week 3 changes (auto concept extraction, question bank) were made behind the facade without touching the CLI dispatch code.
Matches Stage 1?	Yes. Two operations return new result records (`MaterialUpload`, `ActivityOutcome`), see section 4.

3.4 Observer - reacting to progress changes

Design problem	When mastery changes, the dashboard must refresh, the plan may need revision and a review reminder may be due; the tracker must not depend on the GUI, the agent or the scheduler.
Participants and roles	Subject: `ProgressTracker` (`attach()`, `detach()`, `notifyObservers()`). Observer: `ProgressObserver` (`onProgressChanged(ProgressEvent)`). Concrete observers: `ProgressDashboardView`, `StudyPlanMonitor`, `ReminderScheduler`. Event: `ProgressEvent` (old/new mastery, `isSharpDrop()`, `fellBelowWeakThreshold()`).
Files	`application/ProgressTracker.java`, `ProgressObserver.java`, `StudyPlanMonitor.java`, `ReminderScheduler.java`; `gui/ProgressDashboardView.java`
How it follows the pattern	Push model: the event carries old and new mastery. A failing observer is reported and does not stop the others or undo the grading. The monitor and scheduler are attached in `AppContext`; the dashboard attaches itself when created.
Matches Stage 1?	Yes. Difference: the dashboard stays attached for the lifetime of the window rather than only while its tab is visible (simpler, no visible cost).

3.5 Adapter - LLM provider independence

Design problem	Gemini and OpenAI-compatible APIs have incompatible request/response formats, structured-output mechanisms and errors; agent, tools and grading must not depend on either.
Participants and roles	Target: `LLMClient` (`generate()`, `generateStructured()`, `getModelName()`, `isAvailable()`). Adapters: `GeminiLLMAdapter`, `OpenAILLMAdapter`. Adaptees: `GeminiSdkClient` (`generateContent()`), `OpenAISdkClient` (`createChatCompletion()`), thin REST clients over `HttpJson` with the vendors' request shapes. Clients: `LearningAgent`, `Planner`, the generation tools, `GradingService`.
Files	`integration/llm/*`; provider chosen by `LLMClientFactory` from `LLM_PROVIDER`
How it follows the pattern	Adapters translate messages, JSON-mode requests, token counts and errors (into `LLMException`). Switching provider is a `.env` change; all tests use a fake `LLMClient`.
Matches Stage 1?	Yes. Added around it: `LLMClientFactory`, `UsageMeter` (wraps any client to count calls and tokens) and `UnavailableLLMClient` (no key).

3.6 State - adaptive difficulty

Design problem	An adaptive session behaves differently at each level (question types, when to move up or down); `if (level == ...)` checks would scatter the rules.
Participants and roles	Context: `AdaptiveLearningSession` (`recordOutcome()`, `setState()`, `getState()`). State: `DifficultyState` (`getLevel()`, `onResult()`, `preferredQuestionTypes()`). Concrete states: `FoundationState`, `IntermediateState`, `AdvancedState`.
Files	`domain/AdaptiveLearningSession.java`, `DifficultyState.java` and the three states

How it follows the pattern	Each state decides its own transition from the session's streak; the agent asks the current state which question types to generate and the `QuestionBank` fallback ranks stored questions by the same preferences.
Matches Stage 1?	Yes.

3.7 Updated UML class diagrams

The figures below show the implemented system in three views (as in Stage 1). Blue classes were added or changed in Stage 2. The complete specification-level model, with every class, attribute and method, is `docs/stage2/diagrams/class-model-stage2.iuml` (PlantUML, derived from the Stage 1 model); `scripts/render-stage2-diagrams.sh` renders it. Every class of the Stage 1 class model exists in the code; 50 classes were added: 23 enums, exceptions and small helpers, the 12 SQLite persistence classes, and 15 further classes (`ActivityOutcome`, `AppConfig`, `AppContext`, `ConsoleNotificationGateway`, `GeminiEmbeddingService`, `GuiContext`, `LLMClientFactory`, `LocalHashEmbeddingService`, `MaterialUpload`, `QuestionBank`, `QuestionPanel`, `StudyAgentGui`, `StudyCommands`, `UnavailableLLMClient`, `UsageMeter`), all explained in section 4.

Use-case and sequence diagrams. No use case was added or removed, so the Stage 1 use-case diagram remains accurate. The Stage 1 sequence diagrams remain accurate for the main flows; the changes are alternative flows or return types: SD07 (upload) now returns `MaterialUpload` and its "embedding unavailable" alternative keeps the material pending; SD04 (adaptive session) returns `ActivityOutcome` and has the stored-question alternative described in F12. These are documented in section 4 rather than redrawn.

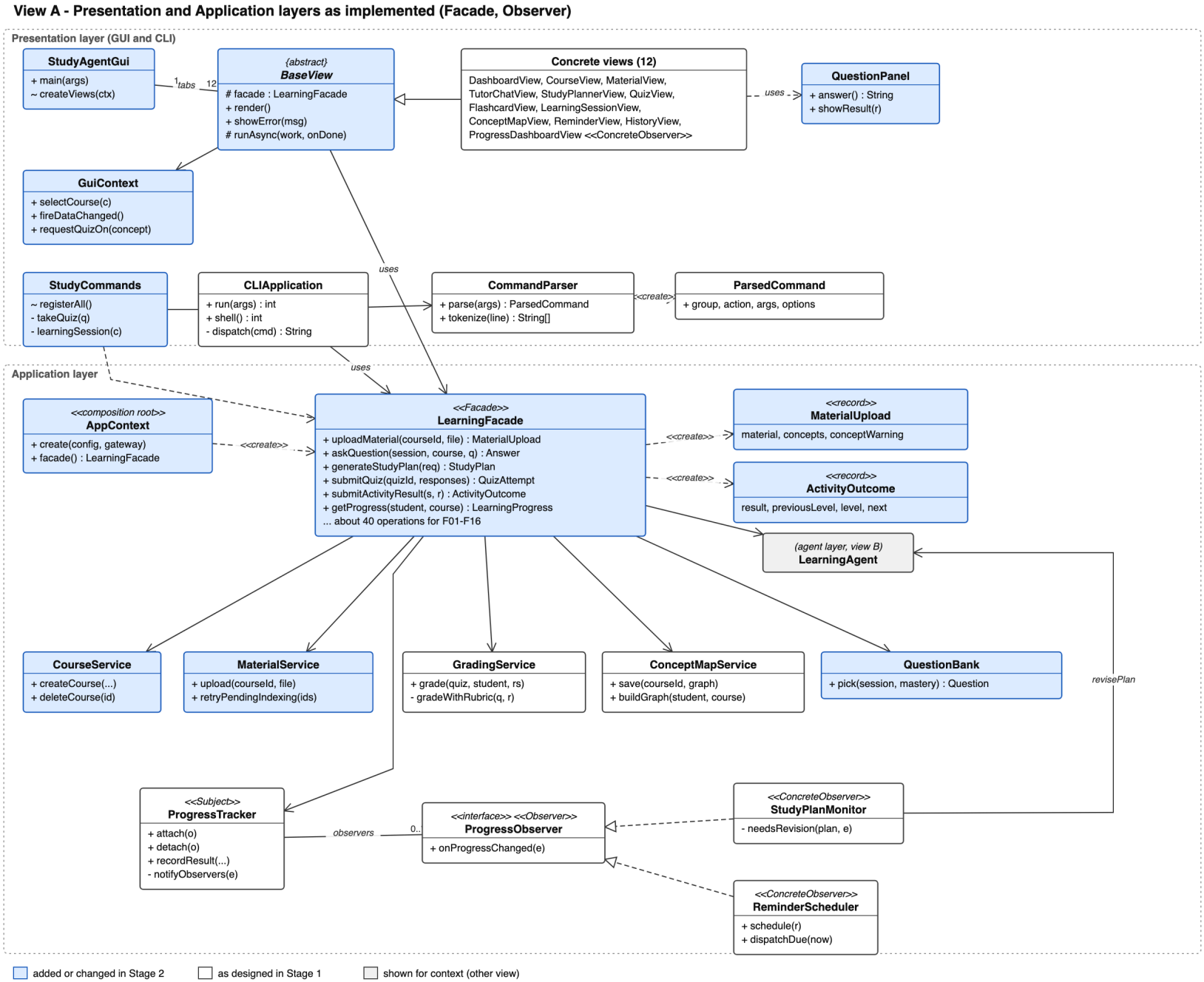


Figure 1. View A - presentation and application layers (Facade, Observer).

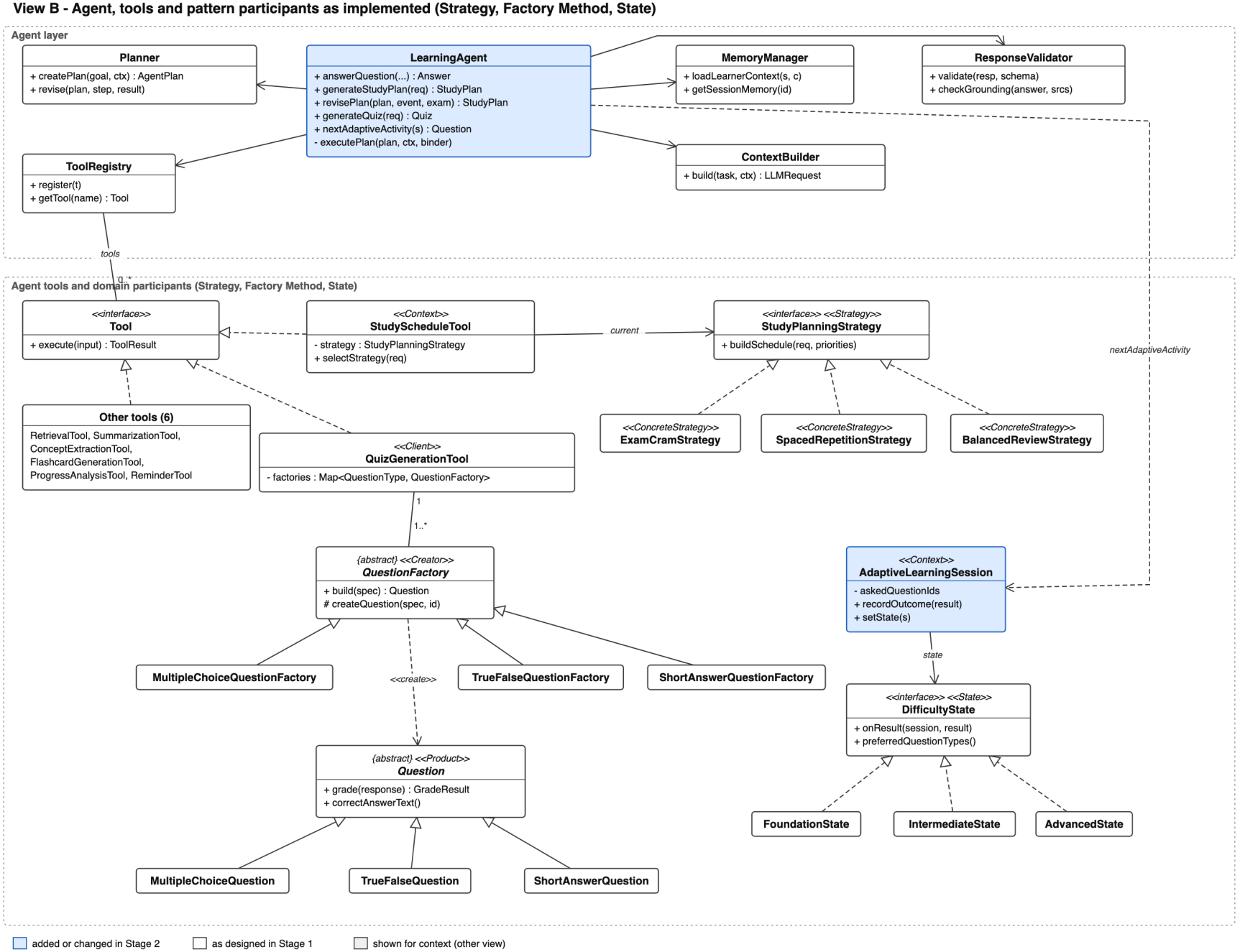


Figure 2. View B - agent, tools and the Strategy, Factory Method and State participants.

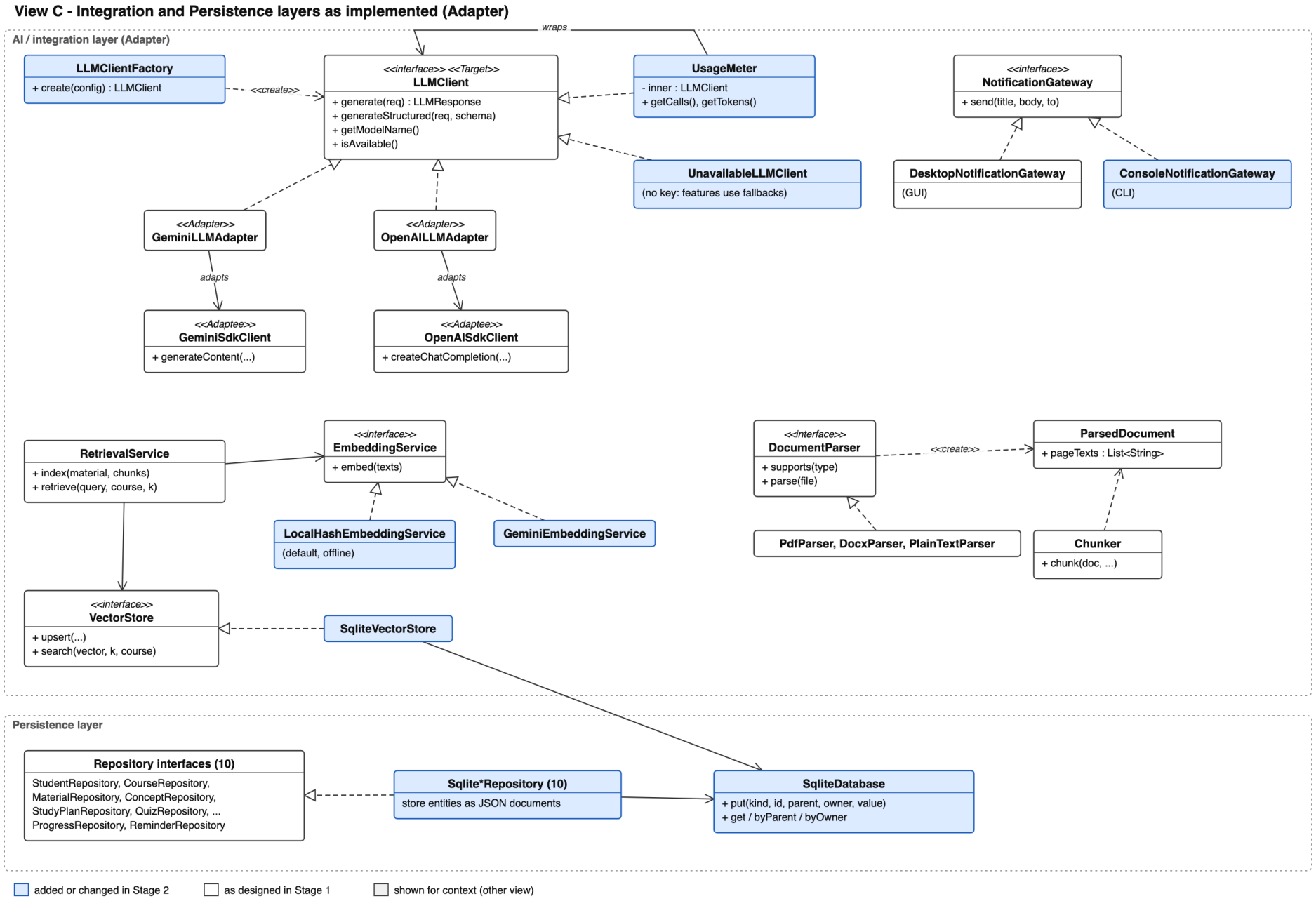


Figure 3. View C - integration (Adapter) and persistence layers.

4. Changes and rationale

Structure: original design → implemented change → reason → impact (affected use cases, classes, methods, patterns). No Stage 1 feature was removed or replaced and no design pattern was replaced.

4.1 Feature changes

#	Original design (Stage 1)	Implemented change	Reason	Impact
C1	F02/F04: UC02 includes UC04; `uploadMaterial()` returns `CourseMaterial`.	Modified F02: after indexing, the facade runs concept extraction and returns `MaterialUpload(material, concepts, conceptWarning)`.	The front ends must tell the student which concepts were found or why not, and an extraction failure must not undo the upload.	UC02, UC04; `LearningFacade.uploadMaterial()`, new `MaterialUpload`, `MaterialView`, CLI `material upload`; Facade.
C2	F12: `submitActivityResult()` returns the next `Question`.	Returns `ActivityOutcome(result, previousLevel, level, next, notice)`; added `endAdaptiveSession()`.	The UI must show the grade and feedback of the answer just given and whether the level changed, which a bare question cannot carry.	UC12; `LearningFacade`, new `ActivityOutcome`, `LearningSessionView`, CLI `learning-session`; State, Facade.
C3	F02: "embedding service unavailable → status remains UPLOADED, indexing retried later" (behaviour, no API given).	`CourseMaterial.markIndexingPending()`, `MaterialService.retryPendingIndexing()`, retry at start-up, CLI `material reindex`, GUI Retry indexing.	Implements the specified alternative flow; the first version discarded the parsed text on any failure.	UC02; `MaterialService`, `CourseMaterial`, `LearningFacade.retryPendingIndexing()`, `CLIApplication`, `StudyAgentGui`, `MaterialView`.
C4	F12: "LLM unavailable → next question drawn from previously generated questions for the same concept and level".	New `QuestionBank`: unused questions of the course's earlier quizzes, ranked by level and preferred type, then least-mastered concept; the session tracks asked ids.	A dedicated class keeps the selection rule deterministic and testable and out of the facade.	UC12; `QuestionBank`, `AdaptiveLearningSession.askedQuestionIds`, `LearningFacade.nextActivity()`; State (uses `preferredQuestionTypes()`).
C5	F05: "LLM unavailable → error message".	The agent returns the three most relevant retrieved passages with citations, clearly labelled.	Retrieval works without an LLM, and showing the student their own material is more useful than an error.	UC05; `LearningAgent.extractiveAnswer()`.
C6	F11: insufficient data below 5 graded attempts.	Implemented first with "a concept has 3 attempts", corrected in week 3 to "5 graded answers in the course"; a concept is weak after 3 attempts below 50%.	Align with the Stage 1 rule.	UC11; `LearningAgent.diagnoseWeakTopics()`.

C7	CLI command list of Stage 1 (Appendix B).	Added `help`, `status`, `material reindex`, `flashcards list`, `quiz list`, `quiz generate --take`, `reminder dispatch`, `study-plan show --all`, `--trace`, `--help`, an interactive shell; `quiz explain` accepts the question number.	Usability and demonstration (the trace shows the agent's plan and tool results).	New class `StudyCommands` (F06-F16 handlers) split from `CLIApplication` to keep each class focused.
C8	GUI: `BaseView` with 12 views.	Swing; added `StudyAgentGui` (window, course selector, tabs), `GuiContext` (selected course, refresh events, "Quiz me on this" requests), `QuestionPanel` (question input shared by quiz and adaptive views); `BaseView.runAsync()` runs LLM calls off the event thread.	Views needed shared state and one way to render questions; LLM calls take seconds and must not freeze the window.	All views; no pattern change.
C9	GUI details: clickable citations; history entries link to objects; status badge.	Citations shown as text; history not clickable; status as a table column.	Time; the information is present.	Listed as limitations (section 1).

4.2 Architecture and class changes

#	Original design	Implemented change	Reason	Impact
A1	Repository interfaces only; "local SQLite".	`SqliteDatabase` stores entities as JSON documents (kind, id, parent, owner) via Gson with `JsonCodec`; ten `Sqlite*Repository` classes; vectors in their own table (`SqliteVectorStore`).	One schema-free table keeps persistence small while the domain model evolves; polymorphic `Question`s are stored with their type.	Persistence layer only; interfaces unchanged.
A2	`EmbeddingService` (provider not fixed).	`LocalHashEmbeddingService` by default (hashed, stemmed bag of words); `GeminiEmbeddingService` with `EMBEDDING_PROVIDER=gemini`. Vectors store their model name.	Works offline, costs no quota and is deterministic for tests; the free Gemini tier is kept for generation.	Retrieval is lexical by default (limitation in section 1).
A3	`LLMClient` with two adapters.	Added `LLMClientFactory`, `UsageMeter` (call and token counter shown by `study status`), `UnavailableLLMClient`.	Configuration-driven provider choice; usage visibility; explicit no-key mode instead of null checks.	Adapter unchanged; integration layer.
A4	Objects created where needed.	`AppContext` composition root and `AppConfig` (environment, `.env`, properties).	The GUI, the CLI and the tests build the identical object graph in one place.	Both `main()` methods; tests use `AppContext.create(config, db, llm, gateway)`.
A5	`DesktopNotificationGateway`.	Added `ConsoleNotificationGateway` for the CLI.	Terminal users see reminders in the terminal.	F14.
A6	Student profile (no login specified).	One local profile created at start-up (`STUDENT_ID`, `STUDENT_NAME`).	Single-user desktop application.	No authentication.

4.3 Design pattern changes

None of the six patterns was replaced or substantially modified; their participants are those of Stage 1 (section 3). Patterns not counted toward the five: the template operation `QuestionFactory.build()` (part of Factory Method), `UsageMeter` wrapping an `LLMClient` (decorator-like), and the CLI's command registry.

5. AI tools and collaboration log

5.1 Tools used

Tool / platform	LLM / model	Used for (features, classes, tasks)	Tokens and cost
[week 1 tool, e.g. ...]	[model]	Week 1 (Oct 26) commits: domain model, persistence, LLM adapters, parsing and retrieval, agent core, strategies, factories, generation tools [confirm]	[tokens / $]
Claude Code (VS Code extension)	Claude Opus 5.5 (`claude-opus-5-5`)	Week 2: `LearningAgent`, application services and `LearningFacade`, `AppContext`, CLI (`CLIApplication`, `CommandParser`, `StudyCommands`), Swing GUI (12 views). Week 3: gap analysis against the Stage 1 specification and fixes C1-C6, F15 "Quiz me on this", sample data, demo script, class diagrams and this report draft. Tests for all of the above.	[tokens / $ from the Claude usage page]
Google Gemini API (runtime, not a development tool)	`gemini-2.5-flash`	The application's own LLM at run time (free tier).	[$0 on the free tier / usage]

5.2 Representative interactions

Request	What the AI did	Human decisions and review
"The professor said Stage 2 must be in Java; change the project to Java."	Inspected the repository and reported that all 162 source files were already Java (Gradle, Java 21), built it and ran the 57 existing tests, and listed what was actually missing: the agent loop, the application layer, the CLI and the GUI.	No conversion was needed; the student decided to proceed with implementing the missing layers.
"Do 10 commits for Stage 2 week 2."	Implemented the missing layers as ten working, tested commits (agent loop; grading and progress; services; observers; facade; CLI x2; GUI x3). The end-to-end facade test exposed two defects that were fixed before committing (whole-course quiz retrieved nothing; its questions did not update mastery); a CLI test exposed that the parser lowercased the first word of questions.	The student required that commits list only her as author and that AI attribution lines be removed from commit messages; this report documents the AI use instead, as the course requires.
"Complete Stage 2 and make it ready for submission."	Compared every F01-F16 specification with the code and implemented the missing alternative flows (C1, C3, C4, C6, F06 grounding warning and retry, F15 "Quiz me on this"); added sample data and a demo script; running the demo without a key exposed a misleading error message, which was fixed.	[describe your own review: what you read, ran and changed]

Problems in AI-generated code caught during development: a reference to a non-existent helper class (compile error, fixed by using the Planner's tool names); a meaningless statement and a button that stayed disabled after a failed LLM call in `TutorChatView` (fixed with an "always" callback in `runAsync`); a redundant expression in `QuestionBank.pick()`; incorrect test expectations (an empty priorities list that the validator correctly rejects; a flashcard due-date assumption). Each was found by compiling, by the tests, or by reviewing the diff before committing.

6. Development and verification summary

6.1 Weekly commit summary

Links: `https://github.com/Nazaninsoleimani/EECS3311/commit/<id>`. Every listed commit changes implementation or test code. Dates are Toronto time.

Period 1 - October 20-26, 2026 (11 commits, minimum 5)

Commit	Date	Change
605beff	2026-10-26	Set up Gradle multi-module build and core domain model
3815bc5	2026-10-26	Add course material, chunk, summary and concept graph domain classes
d0c0baa	2026-10-26	Implement question hierarchy with polymorphic grading and quiz attempts
fba50f6	2026-10-26	Add progress, flashcard, study plan, reminder and history domain rules
2acdea6	2026-10-26	Add repository interfaces and SQLite persistence layer
4388017	2026-10-26	Implement LLM provider adapters (Gemini, OpenAI-compatible) and config
84d7487	2026-10-26	Add document parsing, chunking, embeddings and vector retrieval
9796881	2026-10-26	Add agent core: planner, tool registry, memory, context builder, validator
5ba5ef9	2026-10-26	Implement study scheduling strategies and progress analysis tool
055a8ae	2026-10-26	Add question factories, quiz generation tool and adaptive difficulty states
532d36a	2026-10-26	Add summarization, concept extraction and flashcard generation tools

Period 2 - October 27 - November 2, 2026 (10 commits, minimum 5)

Commit	Date	Change
87a9da9	2026-10-30	Implement LearningAgent loop with grounded answers and plan execution
b781d5b	2026-10-30	Add GradingService and ProgressTracker subject with progress observers
a04de3b	2026-10-30	Add course, material upload and concept map application services
badde08	2026-10-30	Add ReminderScheduler and StudyPlanMonitor progress observers
4e656eb	2026-10-30	Add LearningFacade and AppContext composition root
80f3ff8	2026-10-30	Add CLI command parser and course, material and tutor commands
150c6e1	2026-10-30	Add CLI commands for study plans, practice, progress and reminders
9c4746f	2026-10-30	Add Swing GUI shell with course, material and tutor views
1684648	2026-10-30	Add GUI study planner, quiz, flashcard and adaptive session views
3623aea	2026-10-30	Add GUI dashboard, progress observer, concept map, reminder and history views

Period 3 - November 3-9, 2026 (7 code commits, minimum 5)

Commit	Date	Change
3fc7736	2026-11-05	Extract concepts automatically after a material is uploaded (C1)
bee0f15	2026-11-05	Reuse stored questions when adaptive generation fails (C4)
924acd6	2026-11-05	Keep materials pending and retry indexing after embedding failures (C3)

58cae4e	2026-11-05	Reject past exam dates and require five graded answers for diagnosis (F01, C6)
d8bb629	2026-11-05	Ground study plans in concept material and retry invalid priorities (F06)
7a220df	2026-11-05	Add Quiz me on this from the concept map (F15)
e133221	2026-11-05	Add sample course material and a CLI demo script (and fix a misleading error)

6.2 Checks performed

Check	Result
Build (`./gradlew build`)	Compiles `backend`, `frontend:cli`, `frontend:gui` with Java 21; 118 JUnit tests, 0 failures.
Backend unit and integration tests (87)	Domain rules (grading, mastery, Leitner, reminders, concept graph); SQLite repositories; LLM adapters against canned vendor responses (request shape, response mapping, retry, empty answers); retrieval pipeline; agent core (validator, context builder, planner); strategies, factories and states; generation tools with a scripted LLM; `LearningAgent` (grounding and regeneration, fallbacks, plan priorities and retry, adaptive concept choice, weak-topic thresholds); services (uploads, indexing retry, exam dates, cascade delete, concept-map remapping); observers; adaptive fallback; an end-to-end facade test through all sixteen features.
CLI tests (14)	Parser (options, flags, quoting, invalid values); every command group run against an in-memory backend with scripted input (quiz taking, flashcard review, adaptive session, reminders, history filters, help listing all features, exit codes).
GUI tests (17, headless)	Each view built without a window and checked against facade data: course selection, materials and summaries, tutor answers with sources, plan table, quiz grading display, flashcard flow, adaptive outcome, live progress observer update, concept map details and "Quiz me on this", reminders, history filter, dashboard, all 12 tabs present.
Manual runs without an API key	Real GUI launched against a scratch database (no errors). `scripts/demo-cli.sh` over `sample-data/`: course creation, uploads, cited extractive answers, clear errors for AI-only features, history.
Manual run with the Gemini key	[date, which features you exercised, any problems observed]
API-key protection	`.env` is ignored by git and has never been committed (checked in the history); `.env.example` contains placeholders only; keys are read only by `AppConfig`.

6.3 Important defects found and corrected

Defect	How it was found	Correction
Whole-course quiz retrieved no material (generic query)	End-to-end facade test	Query with the course's concept names; sample chunks across the material as a fallback
Whole-course quiz questions had no concept, so mastery never changed	End-to-end facade test	Generation targets the course's known concepts
CLI lowercased the first word of a question (`study ask What ...`)	Parser test	Lowercase only for command lookup
Tutor "Ask" button stayed disabled after a failed LLM call	Code review	`runAsync` overload with an always-run callback
Any indexing failure discarded the parsed text and marked the material failed	Stage 1 specification review	Indexing-pending state and retry (C3)

Study plan without focus topics searched with an empty query and could not be grounded	Stage 1 specification review	Concept names as query; ungrounded warning
"Upload material first" shown although material was uploaded (no concepts without AI)	Demo script run without a key	Message explains that concepts come from AI extraction

6.4 Remaining limitations

See section 1 (scripted LLM in tests, lexical default retrieval, no OCR, simplified GUI details, single user, variable short-answer grading). Stage 3 will add the full deterministic test plan and KUMA agent validation.

7. Demonstration index

Each segment shows the feature ID and name, how it is invoked and its input, the running result, where AI is involved, and an error or alternative case. Together the recordings show both the GUI and the CLI. No API key appears in any recording. Sample data: `sample-data/EECS3311/`.

Feature	Interface	What the segment shows (incl. error / alternative case)	Video link and timestamp
F01 Course Management	GUI + CLI	Create, edit, delete with confirmation; duplicate code and past exam date rejected	[link @ mm:ss]
F02 Material Upload	GUI + CLI	Upload the three sample notes, status table; `.pptx` rejected; empty file FAILED	[link @ mm:ss]
F03 Summarization	GUI	Brief and detailed summary with key points	[link @ mm:ss]
F04 Concept Extraction	GUI + CLI	Concepts listed after upload; re-extract	[link @ mm:ss]
F05 Question Answering	GUI + CLI `ask -i`	Cited answer, follow-up "why is that?", "Show agent plan"; unrelated question → not covered	[link @ mm:ss]
F06 Study Plan	GUI	Generate with exam date and minutes; strategy and rationale; mark task done; plan revision after a poor quiz	[link @ mm:ss]
F07 Flashcards	CLI + GUI	Generate a deck, review with ratings, next-review dates	[link @ mm:ss]
F08 Quiz Generation	GUI	Choose types, difficulty, concepts; notice when fewer questions pass validation	[link @ mm:ss]
F09 Quiz Grading	GUI + CLI `quiz take`	Submit with an unanswered question; per-question feedback; short answer graded by AI	[link @ mm:ss]
F10 Answer Explanation	GUI + CLI	Explain a wrong answer with sources	[link @ mm:ss]
F11 Weak Topics	GUI + CLI	Insufficient-data message first; after practice, weak topics with AI diagnosis	[link @ mm:ss]
F12 Adaptive Difficulty	CLI + GUI	Three correct answers → level up; two wrong → level down	[link @ mm:ss]
F13 Progress Tracking	GUI + CLI	Progress tab updating live after a quiz (Observer); `progress show`	[link @ mm:ss]
F14 Reminders	GUI + CLI	Add a reminder two minutes ahead and receive the notification; past time rejected; cancel	[link @ mm:ss]
F15 Concept Map	GUI + CLI	Coloured map, click a concept, Quiz me on this; Mermaid output in the CLI	[link @ mm:ss]
F16 History	GUI + CLI	Filters by course, period and activity type; `history show --last 7d --type quiz_completed`	[link @ mm:ss]

Appendix - running the submission

```
cp .env.example .env            # put your own key in .env (never committed)
./gradlew build                 # compile and run the tests
./gradlew :frontend:gui:run     # GUI
./gradlew :frontend:cli:installDist && frontend/cli/build/install/study/bin/study help   # CLI
scripts/demo-cli.sh             # CLI walkthrough over sample-data/ in a separate demo database
```

Full instructions, prerequisites and example usage are in the repository README.